

Quantum chaos

Operators, spectra, and black-hole scrambling

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Quantum chaos is not a single test. Spectral correlations, operator growth, relaxation, and information scrambling probe different aspects of dynamics, often on very different time scales.

This note develops the algebra of out-of-time-order correlators, explains the assumptions of the thermal chaos bound, introduces random-matrix diagnostics and eigenstate thermalization, and follows the connection to black-hole shock waves and the Sachdev–Ye–Kitaev model.

The prerequisites are quantum mechanics, statistical mechanics, and basic field theory. Standard notation for Hamiltonians, thermal density matrices, operators, and Lyapunov exponents follows the cited literature. Units are $\hbar = k_B = 1$ except in the semiclassical comparison, where $\hbar$ is displayed explicitly.

These are pedagogical study notes, with worked calculations and reading references. They summarize established results and indicate the assumptions behind semiclassical or conjectural statements.

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1 Several questions hidden in the word chaos

Classical chaos is usually associated with sensitive dependence on initial conditions in a suitable bounded dynamical system. Quantum time evolution preserves inner products, so exponentially separating state vectors cannot be its direct definition.

Instead, one can ask whether energy levels exhibit correlations described by random-matrix ensembles, whether simple operators spread into complicated many-body operators, whether local observables relax, or whether initially localized information becomes inaccessible to small subsystems. These questions are related in many systems, but none is a universal synonym for the others.

A finite-dimensional unitary system has recurrences and cannot show an exponentially growing bounded observable forever. An exponential regime must be an intermediate-time statement with a specified initial amplitude and saturation scale. The order of large-system, long-time, and low-temperature limits matters.

Integrability supplies further distinctions. A system can have ballistic transport and rapid dephasing without random-matrix spectral statistics. An unstable one-dimensional potential can produce exponential commutator growth without the phase-space mixing characteristic of a bounded chaotic many-body system.

The most reliable study therefore combines observables. Specify the Hamiltonian, symmetries, ensemble, operator choices, and time window. Then test which conclusions are supported by which diagnostic rather than assigning a global label from one attractive plot.

Check. A decaying two-point function does not prove fast scrambling. It measures loss of a particular simple correlation and can decay through ordinary dephasing.

2 Classical sensitivity and its semiclassical image

For a classical trajectory, a small initial displacement can grow as

$$\delta q(t) \sim e^{\lambda_{\text{cl}} t} \delta q(0) \tag{2.1}$$

within a tangent-space regime. The derivative with respect to initial position is a Poisson bracket with the initial conjugate momentum:

$$\{q(t), p(0)\} = \frac{\partial q(t)}{\partial q(0)}. \quad (2.2)$$

A semiclassical commutator therefore behaves as

$$[q(t), p(0)] \simeq i\hbar\{q(t), p(0)\}. \quad (2.3)$$

Squaring it gives growth proportional to $\hbar^2 e^{2\lambda_{\text{cl}} t}$ in the regime where the semiclassical approximation is valid.

The factor of two is not optional. A growth rate extracted from a squared commutator need not equal the classical tangent-space exponent quoted for an unsquared displacement. Different papers sometimes use the same symbol for these different conventions.

The semiclassical approximation fails once initially small quantum uncertainty is stretched to macroscopic scales. This produces an Ehrenfest time logarithmic in the inverse effective Planck constant in appropriate systems. The exact coefficient and physical interpretation depend on the phase-space scales and observable.

A positive local instability exponent also does not guarantee global chaos. Boundedness, repeated stretching and folding, and the absence of enough conserved quantities are part of the classical problem. A single hyperbolic fixed point provides sensitivity without establishing ergodicity.

Check. Before comparing exponents from classical trajectories and thermal four-point functions, identify whether each exponent describes an amplitude, its square, or a connected correction to a normalized correlator.

3 An exactly solvable counterexample

Consider an inverted harmonic oscillator,

$$H = \frac{p^2}{2m} - \frac{1}{2}m\omega^2 q^2. \quad (3.1)$$

The Heisenberg equations are linear and give

$$q(t) = q(0) \cosh(\omega t) + \frac{p(0)}{m\omega} \sinh(\omega t). \quad (3.2)$$

Hence

$$[q(t), p(0)] = i\hbar \cosh(\omega t), \quad (3.3)$$

$$-[q(t), p(0)]^2 = \hbar^2 \cosh^2(\omega t). \quad (3.4)$$

There is exponential growth, yet the equations are exactly linear and the system is not a bounded chaotic many-body system.

The Hamiltonian is unbounded below and does not have the ordinary normalizable Gibbs ensemble needed for the thermal chaos bound. Thus choosing ω larger than an arbitrary temperature scale is not a counterexample to that bound. Its hypotheses are absent.

For comparison, the stable oscillator gives $\cos(\omega t)$ in place of $\cosh(\omega t)$ and an oscillatory squared commutator. The algebraic replacement illustrates how a local instability changes an operator response without creating a many-body spectrum with random-matrix correlations.

This is a valuable diagnostic exercise because it is exact. Any numerical routine for commutator growth should reproduce the corresponding short-time series in a properly controlled truncated basis, while recognizing that truncation itself changes the long-time behavior of an unbounded operator.

Check. The early-time expansion is $\hbar^2[1 + \omega^2 t^2 + O(t^4)]$. A fit to an exponential at arbitrarily small time is inappropriate even in this exponentially unstable example.

4 Operator growth and the squared commutator

For Hermitian operators W and V , the commutator is anti-Hermitian. Therefore

$$C(t) = -\langle [W(t), V]^2 \rangle = \langle [W(t), V]^\dagger [W(t), V] \rangle \geq 0. \quad (4.1)$$

It measures how a perturbation generated by one operator can affect a later measurement of the other. Operator location and normalization are part of the definition.

Expanding the square produces four-point functions with different orderings. At infinite temperature, if W and V are Hermitian unitaries with $W^2 =$

$$V^2 = 1,$$

$$C(t) = 2 - 2 \operatorname{Re} F(t), \quad F(t) = \frac{1}{\dim \mathcal{H}} \operatorname{Tr}[W(t) V W(t) V]. \quad (4.2)$$

This simple identity relies on the stated normalization and trace. At finite temperature, or for nonunitary operators, the time-ordered pieces are not automatically equal to one.

The Heisenberg expansion is

$$W(t) = W + it[H, W] - \frac{t^2}{2}[H, [H, W]] + \cdots. \quad (4.3)$$

If initially $[W, V] = 0$, the first nonzero nested commutator determines the onset of growth. Locality can delay that onset to an order related to the separation of the operators.

In a spin system, expanding $W(t)$ in Pauli strings makes spreading concrete: terms acquire support on more sites. A squared commutator with a local probe measures a weighted portion of that operator support, not the entire operator distribution.

Check. A large commutator with one specially chosen probe need not imply that information has become inaccessible to every small subsystem. Scrambling is a stronger information-theoretic statement.

5 Thermal regularization and ordering

Local field-theory operators can be singular when multiplied at coincident points. A thermal out-of-time-order correlator must include a specified imaginary-time regulator. A common symmetric choice is

$$F(t) = \operatorname{Tr}[\rho^{1/4} V \rho^{1/4} W(t) \rho^{1/4} V \rho^{1/4} W(t)], \quad (5.1)$$

with $\rho = e^{-\beta H} / Z$.

The four quarter-density-matrix factors distribute the insertions around the thermal circle. They are not harmless decorations: the analytic domain and boundedness properties used in the chaos argument depend on this ordering.

The corresponding disconnected normalization is built from regulated two-point functions,

$$F_d = \text{Tr}(\rho^{1/2} V \rho^{1/2} V) \text{Tr}(\rho^{1/2} W \rho^{1/2} W). \quad (5.2)$$

One studies a small departure from this value after ordinary correlators have approximately relaxed but before the out-of-time-order correction becomes order one.

An unregularized thermal squared commutator and the symmetric regulated four-point function need not have identical finite-coupling behavior. A calculation should not move operators around the thermal circle without checking whether the resulting object is the one to which a theorem applies.

For fermionic probes, anticommutators and graded orderings often provide the natural measure of spreading. The signs should be derived from the chosen operator algebra rather than inherited from a bosonic expression.

Reading connection: The regularized ordering used in the thermal chaos bound is specified in Maldacena, Shenker, and Stanford [1].

6 Time scales and a meaningful growth window

The dissipation time describes the relaxation of ordinary simple correlators. The scrambling time marks an order-one out-of-time-order effect for suitably separated or weakly correlated simple operators. A controlled exponential window requires a separation between them.

In a large- N matrix theory, a typical connected correction has the form

$$\frac{F_d - F(t)}{F_d} \sim \frac{1}{N^2} e^{\lambda_L t}. \quad (6.1)$$

Setting the correction to order one gives

$$t_* \sim \frac{1}{\lambda_L} \log N^2, \quad (6.2)$$

up to operator-dependent and coupling-dependent offsets. In a vector-like model such as SYK, the prefactor is often $1/N$, leading instead to $\lambda_L^{-1} \log N$ at leading logarithmic order.

The Heisenberg time is set by the inverse mean level spacing in a specified energy window. In a many-body system it can be exponentially large in

entropy. It is typically much longer than the scrambling time. Spectral plateaus and recurrences therefore probe physics beyond the early growth regime.

A fitted exponent is meaningful only if the signal is above finite-size and numerical errors, below saturation, and stable under reasonable changes in the fit interval. A straight segment on a logarithmic plot lasting less than one e-fold is weak evidence for a parametrically controlled exponential regime.

Check. The logarithm in a scrambling time comes from amplifying a small initial correction. Identify that prefactor before writing $t_* \sim \beta \log N$.

7 The thermal chaos bound and its hypotheses

For an appropriately regulated thermal correlator, analyticity in a complex-time strip and a suitable bound on its magnitude constrain how rapidly it can depart from its factorized value. Under the physical factorization and time-scale assumptions,

$$\lambda_L \leq \frac{2\pi}{\beta} = \frac{2\pi k_B T}{\hbar} \quad (7.1)$$

when constants are restored on the right.

Here λ_L describes the exponential growth of the small connected correction to the regulated four-point function. It is not an upper bound on every quantity called a Lyapunov exponent, every classical orbit instability, or every decay rate.

The mathematical argument maps a bounded analytic function from a thermal half-strip to the unit disk and applies the Schwarz–Pick inequality. The strip width fixes the scale $2\pi/\beta$. To use the result physically, time-ordered correlators must factorize accurately enough that the chosen normalization bounds the analytic function; the correction being measured must exceed the residual factorization error.

Saturation is evidence for a particularly rapid thermal scrambling channel in the relevant limit. It does not determine an entire theory uniquely. Conversely, a numerical value apparently above the bound may reflect a different regulator, no scale separation, an invalid fit window, or hypotheses that do not apply.

Reading connection: Reference [1] states the analytic argument and physical assumptions. The subsequent sections use independent algebraic and gravitational checks to clarify its interpretation.

8 A mathematical check on exponential growth

The bound on an analytic normalized correlator implies, well after the initial edge of the half-strip and within the small-correction regime, a differential restriction of the form

$$\left| \frac{df}{dt} \right| \lesssim \frac{\pi}{\beta} (1 - f^2) \quad (8.1)$$

for a real normalized function near one, with corrections controlled by the half-strip boundary and normalization errors.

Write the small departure explicitly as $f(t) = 1 - \varepsilon e^{\lambda_L t}$. Then

$$1 - f^2 = 2\varepsilon e^{\lambda_L t} - \varepsilon^2 e^{2\lambda_L t}. \quad (8.2)$$

At first order in the departure,

$$\lambda_L \varepsilon e^{\lambda_L t} \lesssim \frac{2\pi}{\beta} \varepsilon e^{\lambda_L t}. \quad (8.3)$$

Canceling the positive small correction gives the stated exponent bound. This elementary substitution explains both the factor of two and why the argument applies before saturation.

The substitution is not the full proof: it assumes the analytic bound has already been established. In particular, it does not justify applying the same inequality to an arbitrary oscillating function or an unbounded coordinate operator. The normalization of f and the size of the omitted errors are essential.

At extremely early times, a generic finite system has a power series in time rather than a pure exponential. At very late times, saturation invalidates the linearization. These two limitations explain why a theorem about a controlled middle regime cannot be tested by fitting the entire time trace to one exponential.

Check. If the measured departure is comparable to the factorization error, canceling it in the last equation is not justified.

9 Black-hole boosts and shock waves

Near a nonextremal horizon, Schwarzschild time translations act as boosts in regular null coordinates. The relative energy of an early infalling perturbation and a later probe grows exponentially with their time separation.

For a horizon with surface gravity $\kappa = 2\pi/\beta$, the center-of-mass scattering energy carries a factor proportional to $e^{2\pi t/\beta}$. In a weak gravitational eikonal regime, the scattering phase is proportional to Newton’s constant times that boosted energy and an impact-parameter-dependent propagator.

In a two-sided black-hole geometry, an early perturbation can be represented by a null shock wave that shifts a horizon coordinate. The resulting change in a two-sided or out-of-time-order correlator becomes large after a logarithmic time even when the original perturbation was small.

For an Einstein-gravity holographic theory, this mechanism gives a leading exponent $2\pi/\beta$ in the relevant channel. Finite string length and higher-derivative effects can alter the result, while quantum corrections and backreaction limit the regime of the linearized shock-wave approximation.

The process is a scattering calculation in a specified state, not evidence that all geodesics diverge with the same exponent. A photon-sphere instability outside a black hole is a different observable and need not obey the thermal many-body bound in the same way.

Reading connection: The shock-wave account of scrambling is developed by Shenker and Stanford [2]. The black-hole note derives the surface-gravity normalization used here.

10 Locality, butterfly fronts, and transport

For local bounded lattice Hamiltonians, a Lieb–Robinson bound limits the growth of commutators outside an effective cone. A schematic form is

$$|[W_x(t), V_0]| \leq \text{constant } e^{-\mu(|x| - v_{\text{LR}}|t|)} \quad (10.1)$$

under the relevant finite-range or sufficiently decaying interaction assumptions.

The butterfly velocity describes the motion of a front in a specified operator-growth diagnostic. It is generally less restrictive than the rigorous Lieb–

Robinson velocity and need not equal it. The front can broaden with time, so a single sharp exponential profile is an approximation.

Near a front, one sometimes uses a form proportional to $e^{\lambda_L(t-|x|/v_B)}$. This should not be assumed valid at all distances and times. Diffusive front broadening, velocity-dependent exponents, and conservation laws can change the profile.

Transport coefficients describe the relaxation of conserved densities. Diffusion follows long-wavelength conservation equations, whereas scrambling concerns operator complexity and information spreading. Relations involving a diffusion constant, a butterfly velocity, and a thermal time can hold in particular models but are not universal identities.

Long-range interactions and unbounded local Hilbert spaces require modified locality bounds. A finite propagation velocity cannot be inferred from the word “local” without checking the Hamiltonian and operator assumptions.

Check. A diffusive charge density can coexist with a ballistically spreading operator front. Conserved information and generic operator support need not move in the same manner.

Reading connection: Random-circuit examples of operator hydrodynamics and front broadening are [7,8].

11 Spectral statistics and symmetry sectors

Let ordered energy levels in a fixed symmetry sector be E_n . Neighboring spacings are $E_{n+1} - E_n$. Comparing their distribution with random-matrix predictions requires removing the slowly varying density of states, a procedure called unfolding.

For uncorrelated levels, the unfolded spacing distribution is Poisson, $P(s) = e^{-s}$. Random-matrix ensembles exhibit level repulsion: the density vanishes as a power of small spacing. The symmetry class is determined by antiunitary symmetries and their action, not simply by whether a matrix happens to be real in one convenient representation.

Mixing independent symmetry sectors can erase level repulsion even when each sector is chaotic. Conversely, a short spectrum or aggressive unfolding

can create misleading agreement with a target curve. Edge levels and exact degeneracies need separate treatment.

A useful local statistic is the ratio of consecutive spacings,

$$r_n = \frac{\min(E_{n+1} - E_n, E_n - E_{n-1})}{\max(E_{n+1} - E_n, E_n - E_{n-1})}. \quad (11.1)$$

It lies between zero and one and is less sensitive to a slowly varying density of states. It does not eliminate the need to resolve symmetries or inspect finite-size effects.

Random-matrix universality describes correlations after an appropriate coarse-graining and within a suitable regime. It is not a claim that all entries of a physical many-body Hamiltonian are independent random variables.

Check. Before plotting a spacing histogram, list every exact commuting symmetry used to block-diagonalize the Hamiltonian. This often matters more than the choice of histogram bins.

12 The spectral form factor

The thermal spectral form factor is built from the partition function continued to complex time:

$$K(\beta, t) = |Z(\beta + it)|^2 = \sum_{m,n} e^{-\beta(E_m + E_n)} e^{it(E_n - E_m)}. \quad (12.1)$$

An ensemble average or a suitable time/energy smoothing is usually needed to expose a smooth dip, ramp, and plateau. An individual finite spectrum has fluctuations.

The disconnected contribution is $|\langle Z(\beta + it) \rangle|^2$. Subtracting it leaves the connected form factor, which isolates spectral correlations more directly. Different normalizations alter the plateau height, so normalized and unnormalized plots cannot be compared without conversion.

For a nondegenerate finite spectrum, the infinite-time average removes the off-diagonal oscillating terms and leaves

$$\overline{K(\beta, t)}^t = Z(2\beta). \quad (12.2)$$

Degeneracies add further surviving terms. With normalization by $Z(\beta)^2$, the plateau is $Z(2\beta)/Z(\beta)^2$.

At infinite temperature for a Hilbert space of dimension D , the unnormalized plateau is D and the normalized one is $1/D$. This is an exact finite-spectrum check, independent of whether the system is chaotic. The ramp, rather than the existence of a plateau alone, contains characteristic correlated spectral information.

Reading connection: The relation between black-hole spectral behavior, SYK, and random matrices is studied in Cotler et al. [4].

13 Eigenstate thermalization and dephasing

An initial pure state can be expanded in energy eigenstates. The expectation value of a simple observable contains diagonal terms plus oscillating off-diagonal terms. Under suitable nondegeneracy and energy-width assumptions, dephasing suppresses the off-diagonal contribution for most late times.

Dephasing alone does not make the diagonal ensemble thermal. The eigenstate thermalization hypothesis supplies the additional claim that simple observables have smooth energy dependence in individual eigenstates, with sufficiently small fluctuations and appropriate off-diagonal structure.

If the initial energy distribution is narrow and the diagonal matrix elements vary smoothly over it, the diagonal ensemble agrees with a microcanonical prediction for those observables. Canonical equivalence then requires its own thermodynamic assumptions.

A system satisfying ETH for a selected set of few-body observables can remain in a globally pure state. Thermalization is local or observable dependent. It does not mean that exact unitary evolution has become a stochastic map on the complete Hilbert space.

Integrable systems have additional conserved quantities that can prevent the ordinary Gibbs ensemble from capturing late-time local data. Many-body localization and quantum scars provide further reasons that blanket statements about every eigenstate or every initial condition can fail.

Check. A small long-time fluctuation is not enough to establish ETH. Compare the late-time value with the appropriate ensemble and inspect

how eigenstate-to-eigenstate variation scales with system size.

Reading connection: Srednicki [5] gives an influential formulation of eigenstate thermalization in a concrete chaotic setting.

14 The Sachdev–Ye–Kitaev model

The SYK model contains N Majorana fermions with random all-to-all interactions involving a fixed even number q of fermions. A conventional Hamiltonian is

$$H = i^{q/2} \sum_{i_1 < \dots < i_q} J_{i_1 \dots i_q} \chi_{i_1} \cdots \chi_{i_q}, \quad (14.1)$$

with $\{\chi_i, \chi_j\} = \delta_{ij}$ and Gaussian coupling variance

$$\overline{J_{i_1 \dots i_q}^2} = \frac{(q-1)! J^2}{N^{q-1}}. \quad (14.2)$$

The phase ensures Hermiticity in this Majorana convention. Different normalizations of the fermions require corresponding changes in the variance.

At large N , disorder-averaged two-point functions are governed by a closed set of Schwinger–Dyson equations. This tractability makes SYK a useful laboratory for strong coupling, approximate conformal symmetry, and scrambling.

The all-to-all interaction means the zero-dimensional model has no ordinary spatial butterfly velocity. Spatial chains of coupled SYK-like sites introduce additional transport questions, but those are different models.

At finite N , each disorder realization is an ordinary quantum Hamiltonian with its own discrete spectrum. The disorder average does not imply that a single realization has a mixed quantum state. It is an averaging operation over Hamiltonians, distinct from a thermal trace within one Hamiltonian.

Reading connection: Maldacena and Stanford [3] analyze the large- N solution and its four-point function. The original spin-glass ancestry is Sachdev and Ye [6].

15 Schwinger–Dyson equations and the infrared limit

In standard Euclidean conventions, the large- N saddle satisfies

$$G(i\omega_n)^{-1} = -i\omega_n - \Sigma(i\omega_n), \quad \Sigma(\tau) = J^2 G(\tau)^{q-1}. \quad (15.1)$$

The sign of the Fourier-frequency term depends on the convention for the Euclidean Green function; the time-domain antiperiodicity and ultraviolet normalization must be checked consistently.

At long times and strong coupling, the derivative term becomes subleading. A scale-invariant solution has

$$G(\tau) \propto \frac{\text{sgn } \tau}{|J\tau|^{2/q}}. \quad (15.2)$$

The fermion dimension is $1/q$. A scaling check follows directly from the convolution of G and Σ : their powers must combine with the integration measure to reproduce a delta function.

At nonzero temperature, the conformal two-point function on the thermal circle replaces the power of time by a power of $\sin(\pi\tau/\beta)$, with antiperiodic continuation. This is an infrared approximation; it fails at separations of order $1/J$ where the ultraviolet kinetic term is important.

The limits $N \rightarrow \infty$, $\beta J \rightarrow \infty$, and very long time do not commute freely. A finite system retains level discreteness and eventually leaves the continuum saddle description. A residual entropy obtained in a large- N low-temperature limit should not automatically be read as an exact finite- N ground-state degeneracy.

Check. The conformal scaling is not a solution at arbitrarily short time. The exact Majorana discontinuity at coincident Euclidean times fixes ultraviolet normalization.

16 Four-point ladders and the soft mode

The connected SYK four-point function is obtained at leading order in $1/N$ by summing ladder diagrams. Each rung contains two interaction-dressed propagators and a factor proportional to $(q-1)J^2 G^{q-2}$. The growth

exponent comes from an eigenvalue condition for the appropriate retarded kernel.

It cannot be read off from the two-point scaling dimension alone. A theory can have a similar-looking infrared two-point power law and a different four-point kernel. Chaos is a dynamical statement about interactions and ordering.

In the strong-coupling limit, the conformal saddle has an approximate reparametrization symmetry. The ultraviolet kinetic term breaks this symmetry and gives dynamics to a soft mode. Its effective action involves the Schwarzian derivative,

$$\{f, \tau\} = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2. \quad (16.1)$$

At finite temperature the corresponding action uses the thermal reparametrization, commonly written through $\tan(\pi f/\beta)$.

The soft sector controls enhanced low-energy four-point effects and yields the maximal thermal exponent in the strict strong-coupling, large- N regime of the standard model. Finite coupling lowers the exponent in that model, while finite N limits the available growth window.

The low-energy relation to nearly-AdS₂ gravity is powerful but should not be inflated into a complete higher-dimensional string construction for every SYK realization. The Schwarzian sector is an effective description with a controlled range of energies.

Check. A pole or large contribution in a Euclidean four-point function is not automatically a real-time growth exponent. The correct analytic continuation and retarded ladder kernel are required.

17 A minimal numerical protocol

For a finite spin chain, first construct a Hermitian Hamiltonian and identify exact symmetries. Diagonalize within each symmetry sector for spectral statistics. Keep the full operator representation, or carefully track sector-changing matrix elements, when computing a correlator whose probes do not preserve those sectors.

In the energy basis,

$$W(t)_{mn} = e^{i(E_m - E_n)t} W_{mn}. \quad (17.1)$$

This avoids repeatedly exponentiating the Hamiltonian. Construct the required ordered products and thermal weights explicitly. For a symmetric regulated correlator, use the four quarter-density-matrix factors rather than replacing them by one Boltzmann weight at the beginning.

Useful checks are Hermiticity, $\text{Tr } \rho = 1$, the exact $t = 0$ commutator, positivity of the squared-commutator expectation, and agreement between direct matrix evolution and the energy-basis expression for a small system.

For spectral statistics, remove exact degeneracies by resolving symmetries and restrict to a central energy window before unfolding. For time traces, compare several system sizes and report the fit interval, saturation scale, and numerical precision.

A successful computation need not display a clean exponential. Small systems often have no parametrically wide separation between relaxation and saturation. Reporting that absence is more informative than forcing an exponent from a narrow crossover.

Check. A probe that changes a conserved charge connects different charge sectors. Keeping only one sector for its matrix elements can accidentally set the correlator to zero or compute the wrong observable.

18 Worked example: one spin has no chaos

Take a single spin with

$$H = \frac{\omega}{2} \sigma_z, \quad W = V = \sigma_x. \quad (18.1)$$

Heisenberg evolution gives

$$W(t) = \sigma_x \cos(\omega t) - \sigma_y \sin(\omega t). \quad (18.2)$$

The Pauli algebra then yields

$$[W(t), V] = 2i\sigma_z \sin(\omega t), \quad (18.3)$$

so at infinite temperature

$$C(t) = 4 \sin^2(\omega t), \quad F(t) = \cos(2\omega t). \quad (18.4)$$

These obey $C = 2 - 2 \text{Re } F$ exactly. The early-time growth is quadratic, and the signal is periodic rather than scrambling irreversibly.

The normalized infinite-temperature spectral form factor is

$$\frac{|Z(it)|^2}{4} = \cos^2(\omega t/2). \quad (18.5)$$

Its long-time average is $1/2$, the expected inverse Hilbert-space dimension. A plateau in a time average therefore does not, by itself, certify chaos.

This example supplies an exact unit test for matrix-based calculations without introducing a phenomenological model. It checks the sign in Heisenberg evolution, the commutator normalization, the factor of two in the OTOC relation, and the distinction between oscillatory finite-system behavior and a controlled many-body growth regime.

Check. The squared commutator ranges from zero to four here. Assuming its saturation value must always be one would amount to silently changing the operator normalization.

19 Worked example: ratios for uncorrelated levels

For two independent exponentially distributed spacings with unit mean, the ratio of the smaller to the larger lies in $[0, 1]$. Its density is

$$P(r) = \frac{2}{(1+r)^2}. \quad (19.1)$$

To derive it, choose which spacing is larger, giving a factor of two, then write the smaller spacing as r times the larger. The Jacobian contributes the larger spacing, whose integral is

$$2 \int_0^\infty ds s e^{-(1+r)s} = \frac{2}{(1+r)^2}. \quad (19.2)$$

The mean is

$$\langle r \rangle = \int_0^1 dr \frac{2r}{(1+r)^2} = 2 \log 2 - 1 \simeq 0.386294. \quad (19.3)$$

This provides an exact reference value for the ideal Poisson process, with no unfolding required.

A physical integrable spectrum need not reproduce this ideal distribution at every small system size or special arithmetic point. Exact degeneracies and additional structure can change the result. Similarly, a chaotic spectrum should be compared with the random-matrix class appropriate to its resolved symmetries.

The ratio statistic is useful because it divides nearby spacings, reducing sensitivity to a slowly changing local density. It still becomes unreliable at spectral edges, across abrupt density changes, or when levels from independent sectors are mixed.

Check. The integral of $P(r)$ on $[0, 1]$ is one. Omitting the factor of two in the derivation gives a density with the wrong normalization.

20 Worked example: extracting a scrambling time

Suppose a normalized connected correction in a matrix large- N theory is $N^{-2}e^{2\pi t/\beta}$ over an intermediate window. At the time when it reaches a fixed small threshold δ ,

$$t = \frac{\beta}{2\pi} \log(\delta N^2). \quad (20.1)$$

Changing the threshold shifts the answer by a constant of order the thermal time, while the leading $\log N^2$ dependence remains.

A threshold much smaller than the factorization or numerical error is meaningless. A threshold near one approaches saturation, where nonlinear corrections spoil the leading expression. The useful interval lies between those two limits.

For a vector-like correction proportional to N^{-1} , the same calculation gives $\beta \log N/(2\pi)$ at leading order. Writing both answers as $\beta \log S$ can be useful in a particular model, but the entropy and prefactor relation must first be established rather than assumed.

This calculation also separates a growth rate from a total scrambling time. Two systems with the same maximal exponent can have different initial connected amplitudes and therefore different times to reach a given threshold.

In a spatial system, an additional propagation delay depends on operator separation. A logarithmic dependence on local degrees of freedom and a ballistic spatial delay can both be present, but their combination is model dependent.

Check. The exponent determines the slope of the logarithmic signal; the prefactor determines its horizontal offset. A fit that allows neither offset nor uncertainty can bias the inferred exponent.

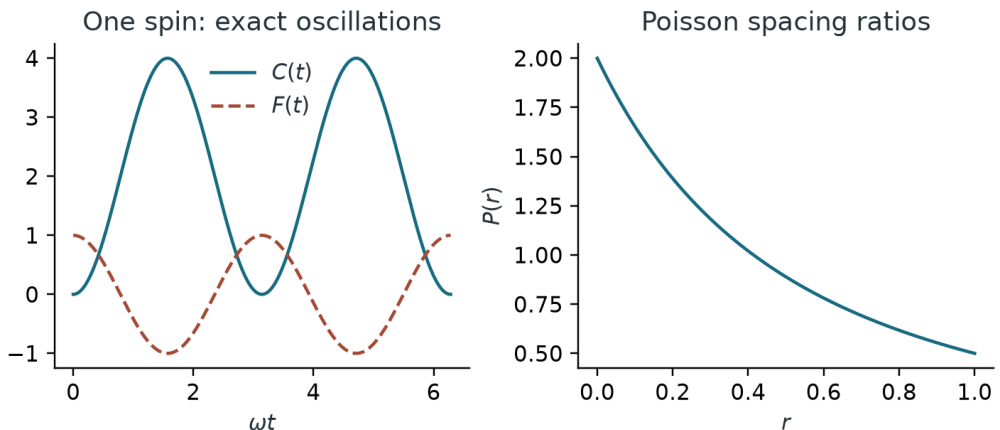


Figure 1: Exact reference curves. A single spin has an oscillating squared commutator and OTOC, rather than a many-body scrambling regime. The Poisson spacing-ratio density supplies a separate spectral benchmark. Neither curve is fitted to numerical data.

21 A hierarchy of conclusions

Spectral level repulsion supports a statement about energy-level correlations in a resolved sector. ETH supports thermal predictions for a class of observables and initial states. A growing squared commutator supports operator sensitivity. A regulated exponential OTOC in a controlled window supports a Lyapunov exponent of that correlator.

Fast scrambling additionally concerns how information becomes inaccessible to simple or small subsystems. A thermal chaos bound constrains one class of rates under analytic and factorization assumptions. A black-hole shock-wave calculation gives a gravitational mechanism for saturating that rate in a particular limit.

References and further reading

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